Statistical Case Studies, Project 1 Part 1, Code

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Executive Summary

Data

- Description of data

Filtering and Cleaning

Table 1: Table 1 — quality codes and their frequency in the data

Code number	Code name	Abbreviated code name	Count	Percentage
50	Good	G	336,175	95.89%
100	Estimated	E	14,388	4.10%
140	Provisional	Prov	0	0.00%
150	Suspect	S	0	0.00%
200	Unchecked (imported from legacy database)	V	0	0.00%
254	Unchecked	U	30	0.01%

- Justify why kept 100 and above but not 254

Exploratory Analysis

Throughout the report, we add three time variables to support our analysis: the month, the season, and the water year. The water year runs from September to August and is labelled by the year it ends in, which keeps each winter's high flows together in the same year.

Main time series and summary table

We first construct the hydrograph for Don at Parkhill over our chosen period.

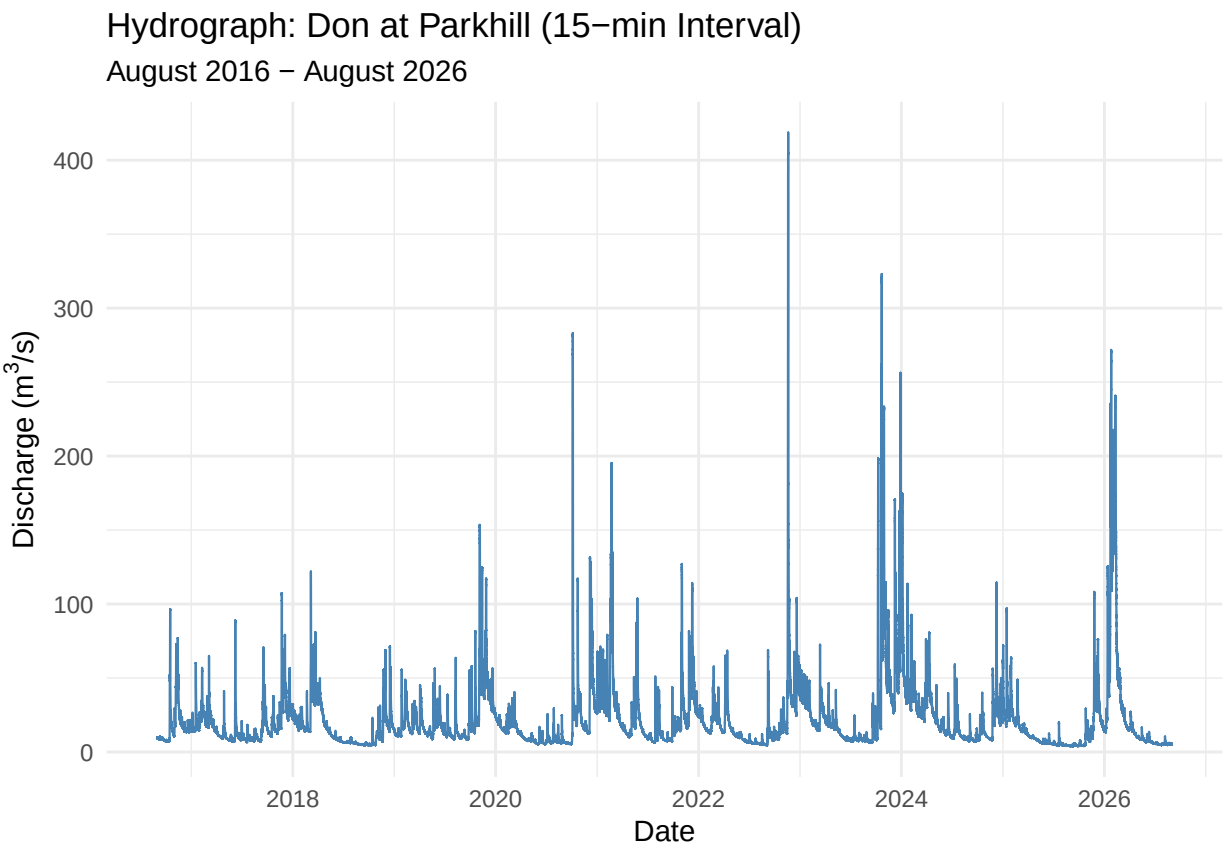
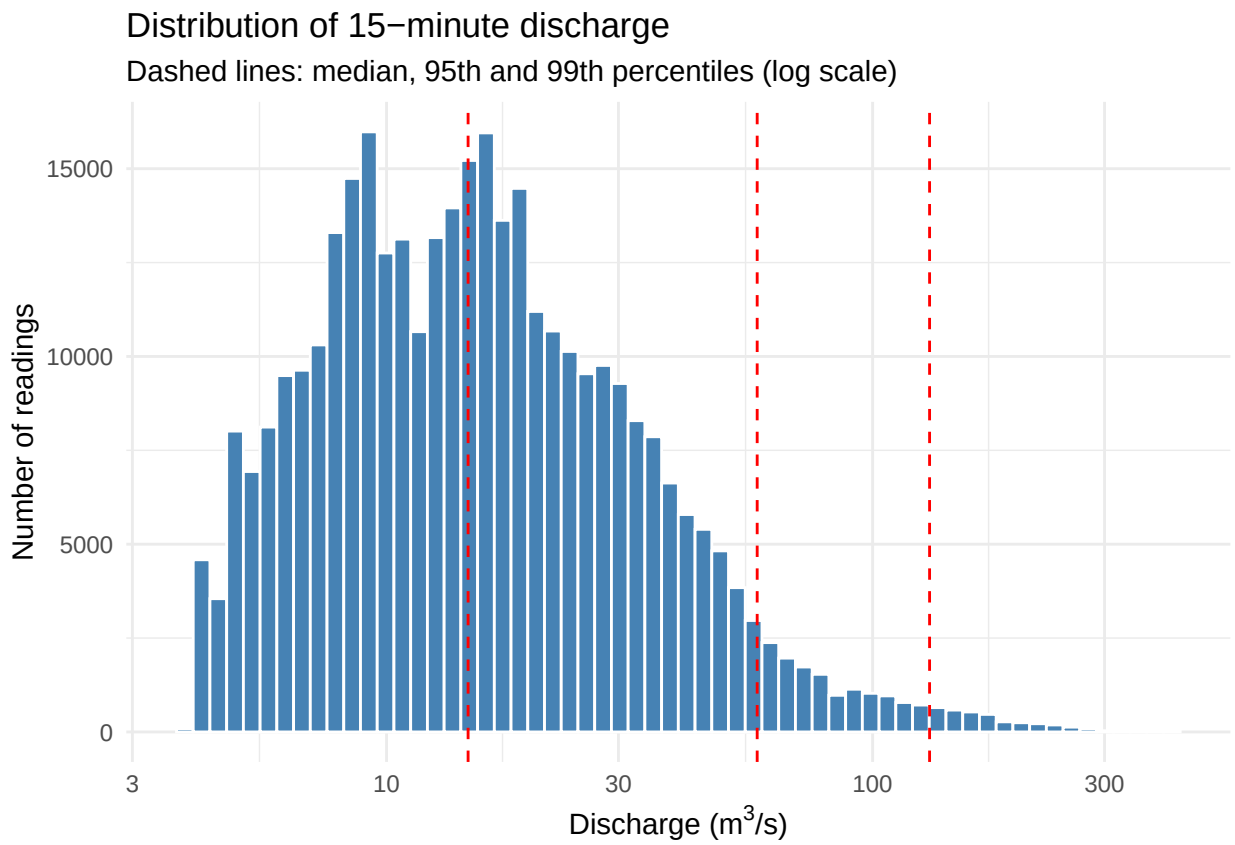


Table 2: Summary statistics for 15-minute discharge (m^3/s)

Statistic	Value
Mean	21.67
Standard deviation	24.37
Minimum	3.90
25th percentile	8.79
Median	14.71
75th percentile	25.44
95th percentile	57.88
99th percentile	130.96
Maximum	418.68



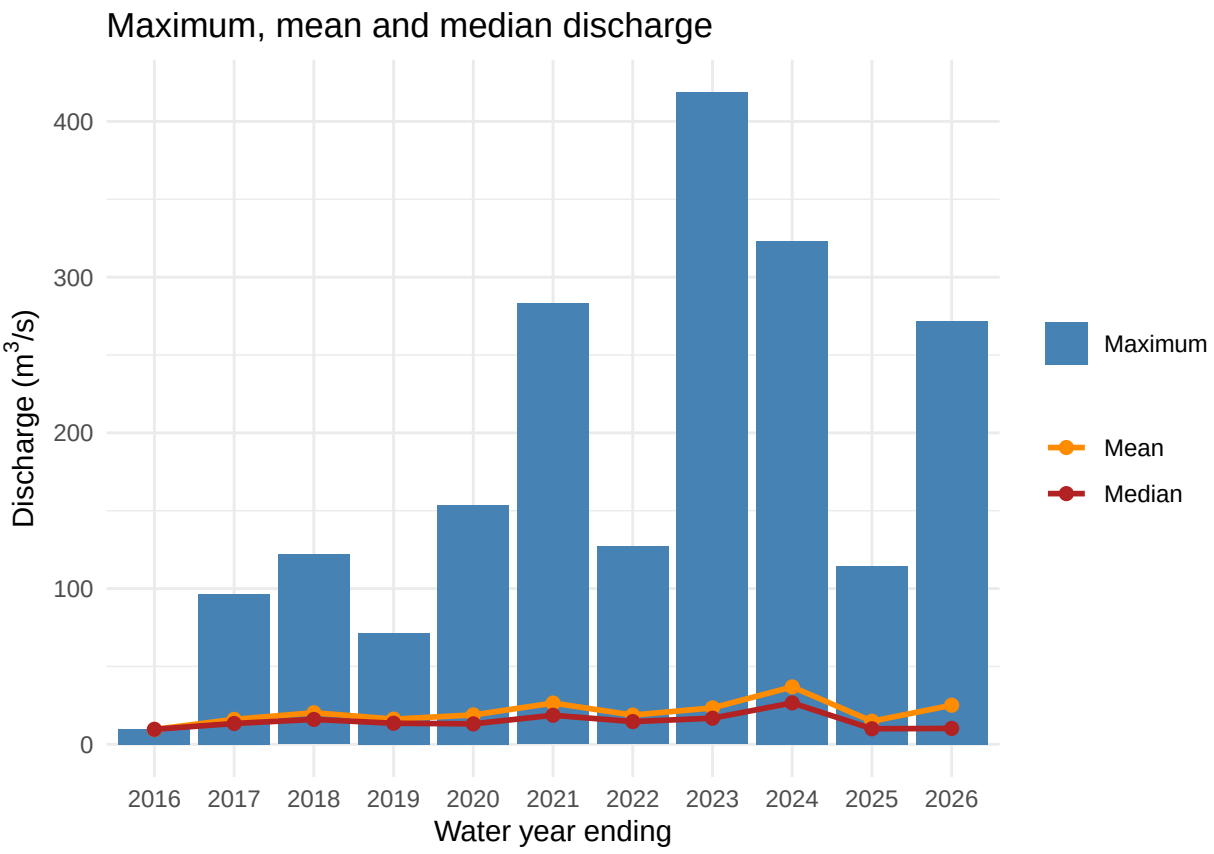
We observe that discharge is low for most of the year, with a smaller number of large peaks. The strong-right skew of the distribution confirms this, with a median discharge of 14.91 m^3/s, and a 99th percentile of 130.96 m^3/s.

Yearly trends

From figure ..., we also observe an upward trend in discharge, and larger maximum flows, over the time period.

Table 3: Discharge by water year (m^3/s)

Water year	Mean	Median	Maximum	Date of maximum
2016	9.54	9.61	9.74	31 Aug 2016
2017	16.01	13.37	96.65	16 Oct 2016
2018	20.24	15.98	121.98	07 Mar 2018
2019	16.18	13.52	71.30	16 Dec 2018
2020	18.86	13.12	153.53	05 Nov 2019
2021	26.48	18.66	283.14	05 Oct 2020
2022	18.75	14.59	127.10	01 Nov 2021
2023	23.39	16.77	418.68	19 Nov 2022
2024	36.86	26.59	323.07	22 Oct 2023
2025	14.83	9.97	114.55	08 Dec 2024
2026	25.09	10.20	271.84	25 Jan 2026


Maximum, mean and median discharge
Discharge (m3/s)
400
300
200
100
0
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Water year ending
Maximum
Mean
Median

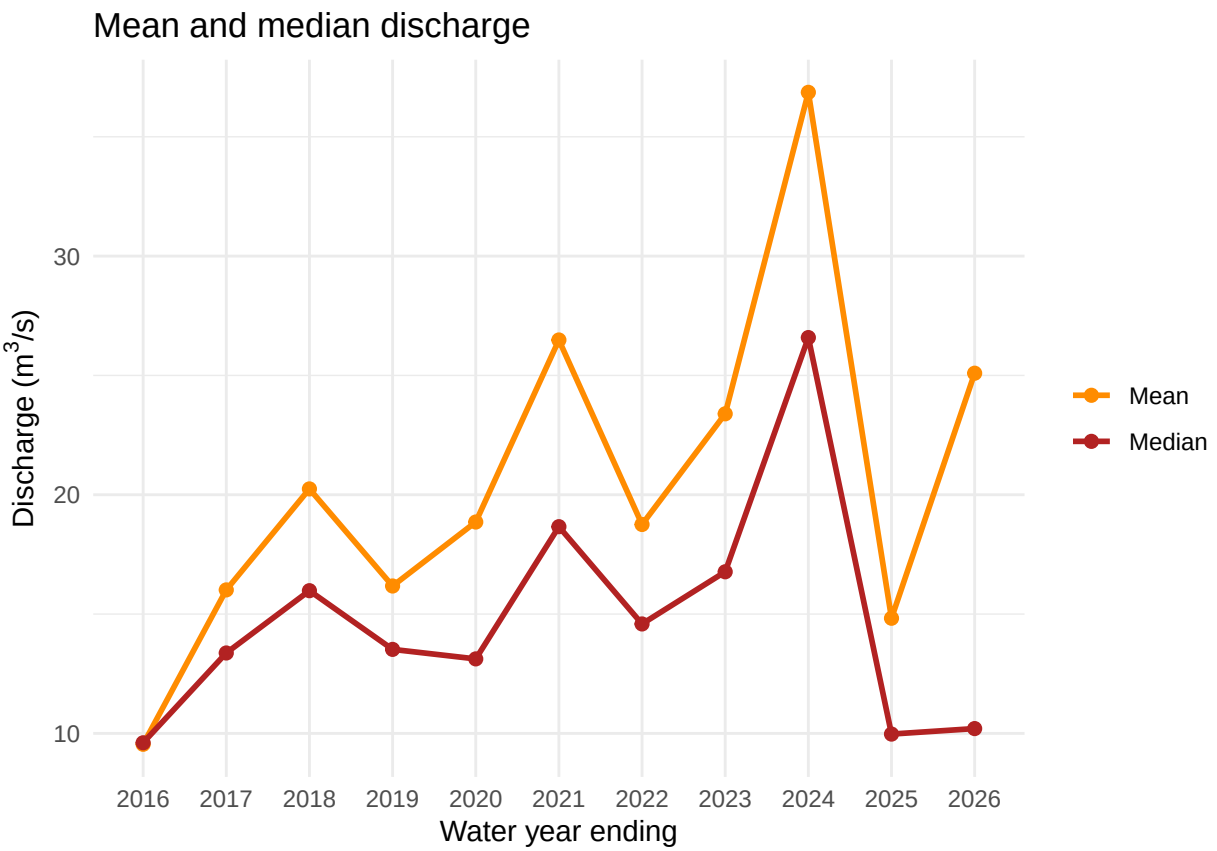


Figure ... provides evidence of this observation, the mean discharge trending upwards from the water year ending in 2017 at ..., peaking in 2024 at .... The maximum discharge follows a similar trend, with both falling sharply in 2025 before rising again.

Exploring seasonal trends overall

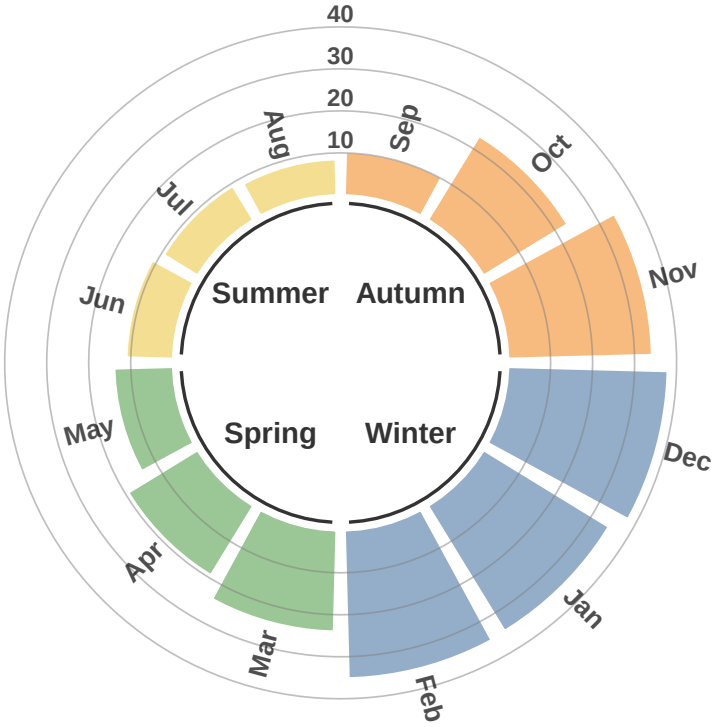
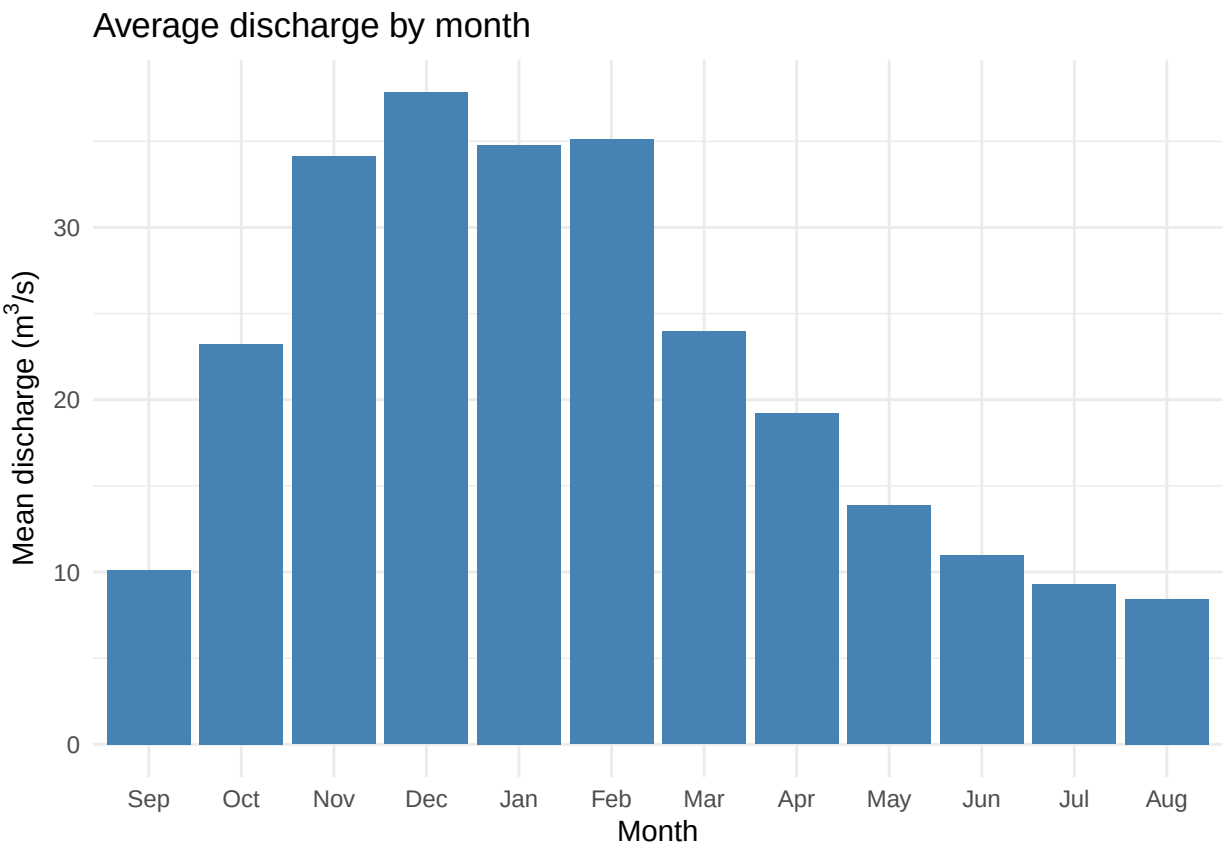
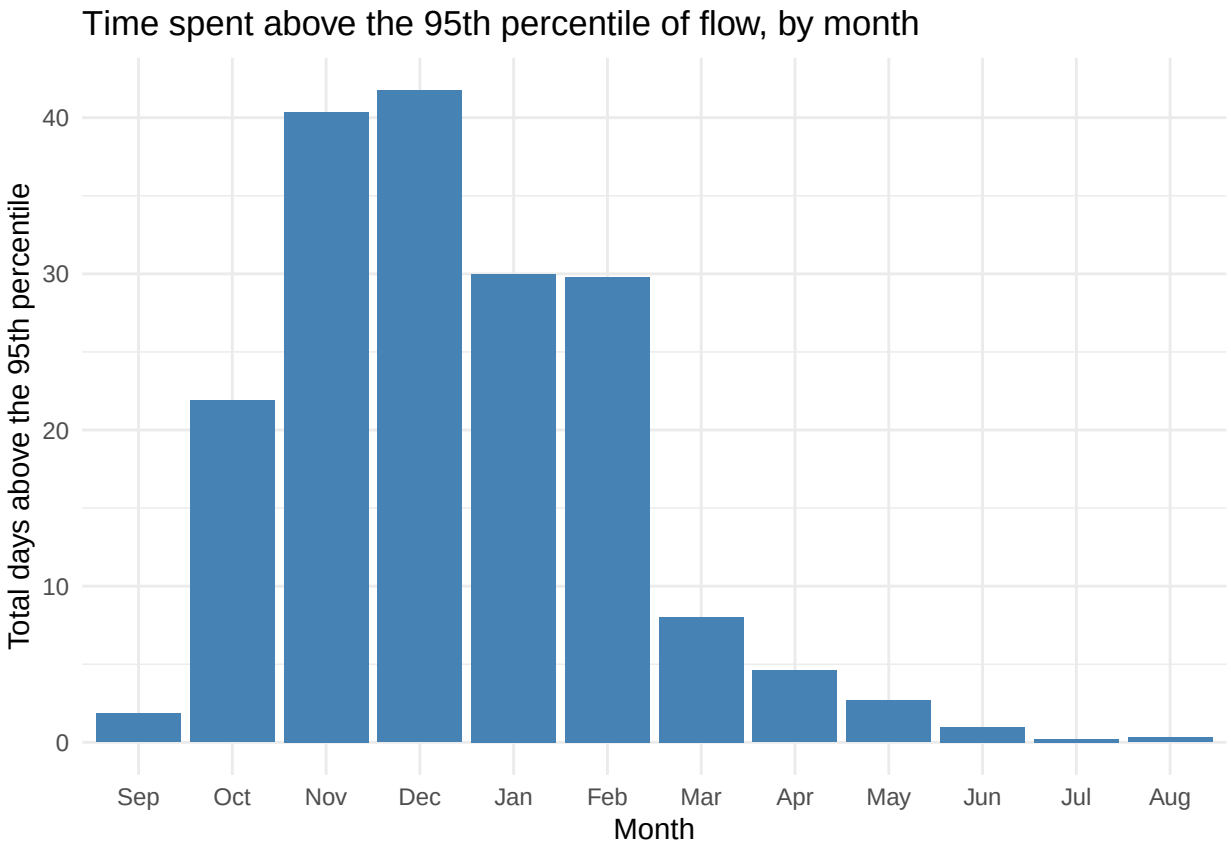
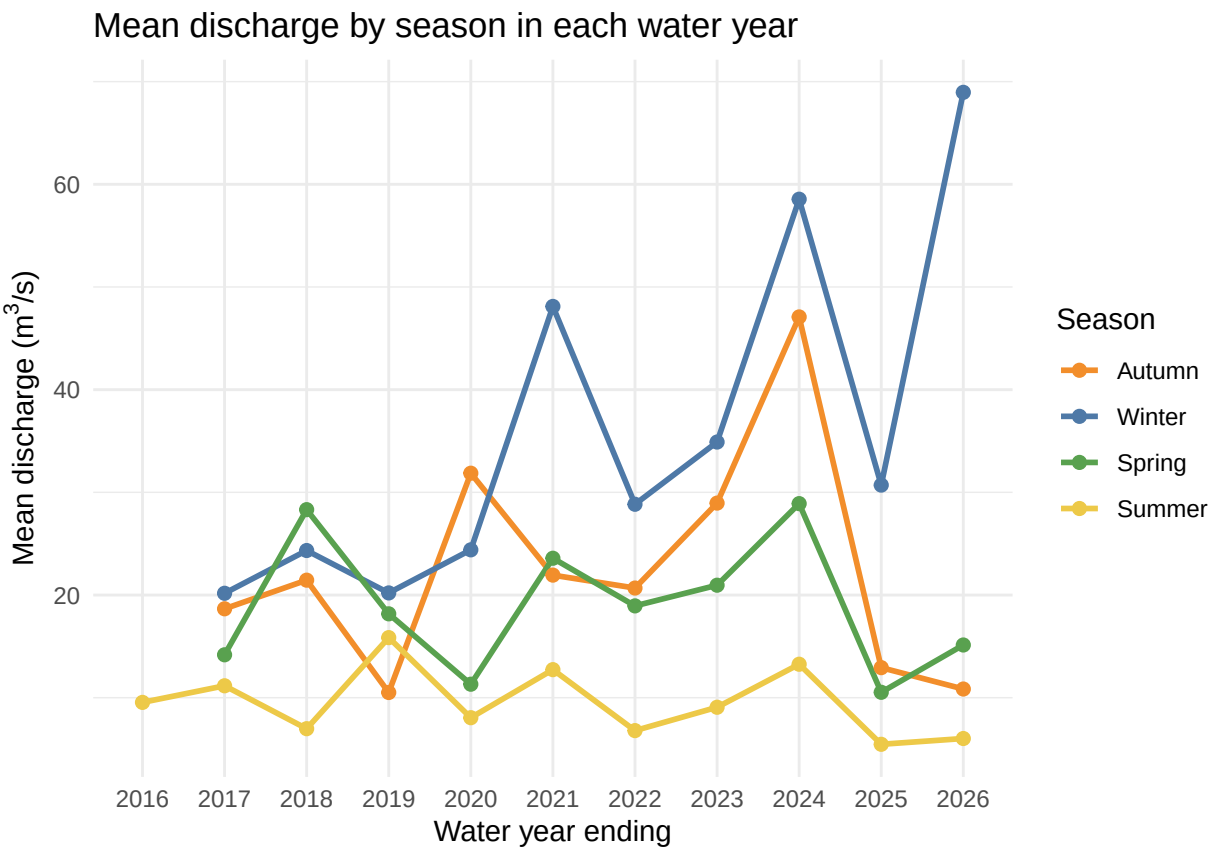


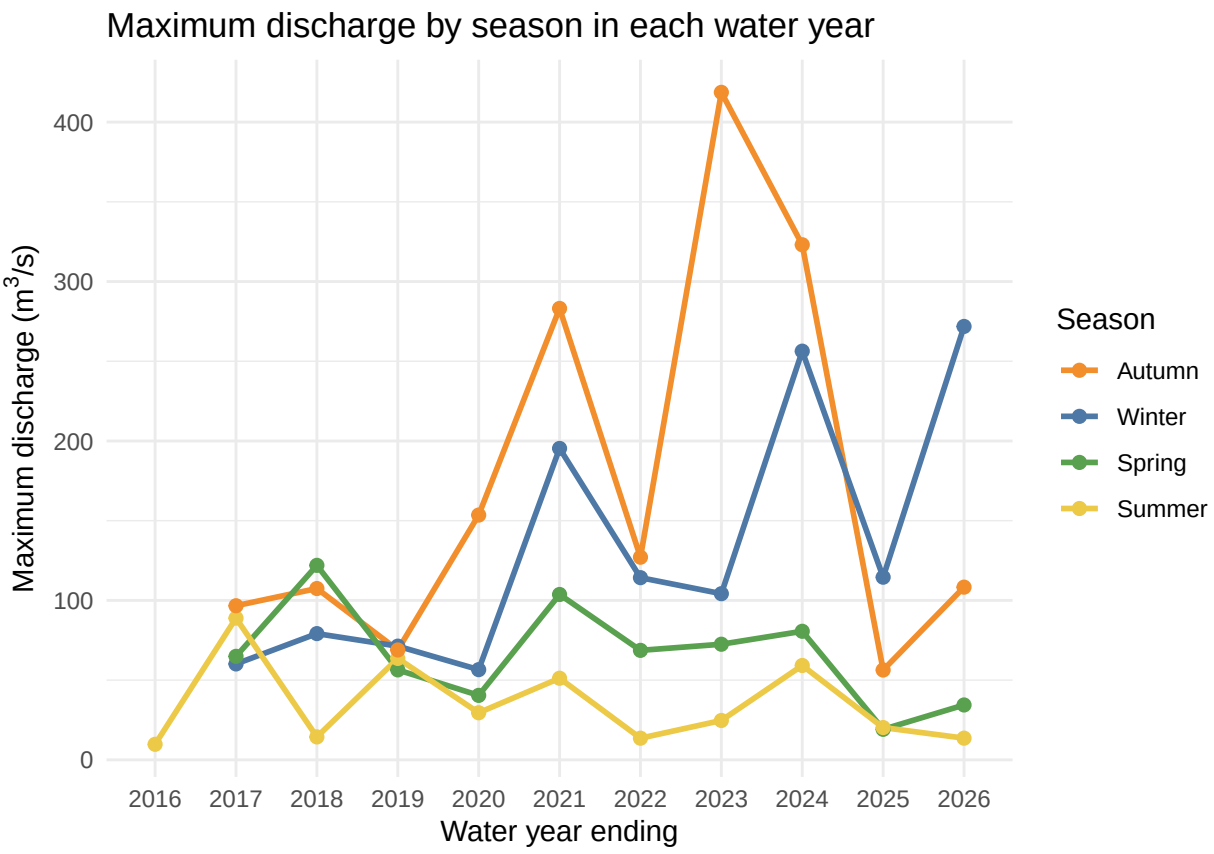
Table 4: Discharge by month (m^3/s)

Month	Mean	Median	95th percentile	Maximum
Sep	10.12	7.91	25.34	70.53
Oct	23.22	14.49	77.46	323.07
Nov	34.11	22.40	87.85	418.68
Dec	37.84	30.60	90.53	256.33
Jan	34.78	24.78	97.58	271.84
Feb	35.09	23.90	122.03	240.84
Mar	23.95	20.55	44.76	121.98
Apr	19.20	15.31	41.15	80.63
May	13.86	11.14	29.56	103.72
Jun	10.96	8.85	22.97	88.89
Jul	9.28	7.75	18.49	59.17
Aug	8.43	7.79	14.83	63.62

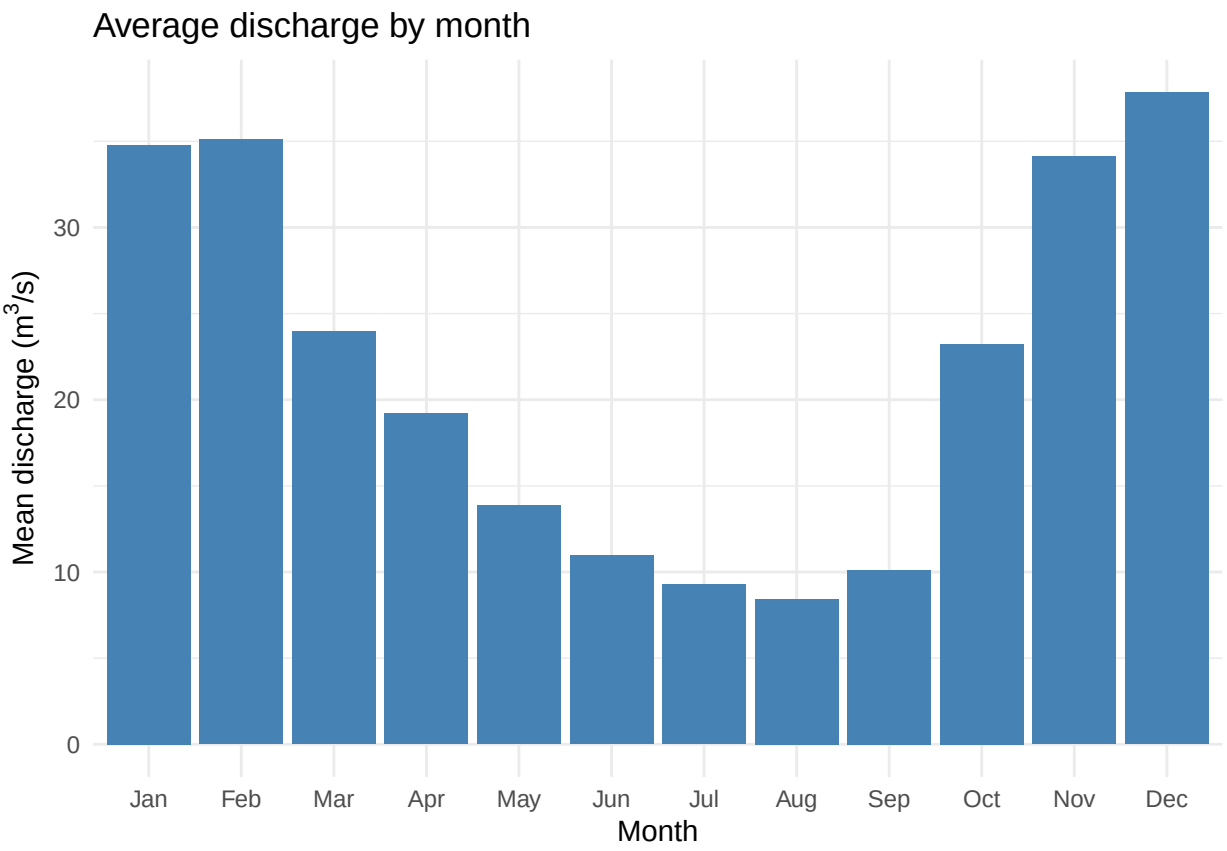

Average discharge by month
Mean discharge (m3/s)
0
10
20
30
Sep
Oct
Nov
Dec
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Month


Time spent above the 95th percentile of flow, by month
Total days above the 95th percentile
0
10
20
30
40
Sep
Oct
Nov
Dec
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Month


Mean discharge by season in each water year
Mean discharge (m3/s)
60
40
20
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Water year ending
Season
Autumn
Winter
Spring
Summer



- Other seasonality options.


Average discharge by month
Mean discharge (m3/s)
0
10
20
30
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Month

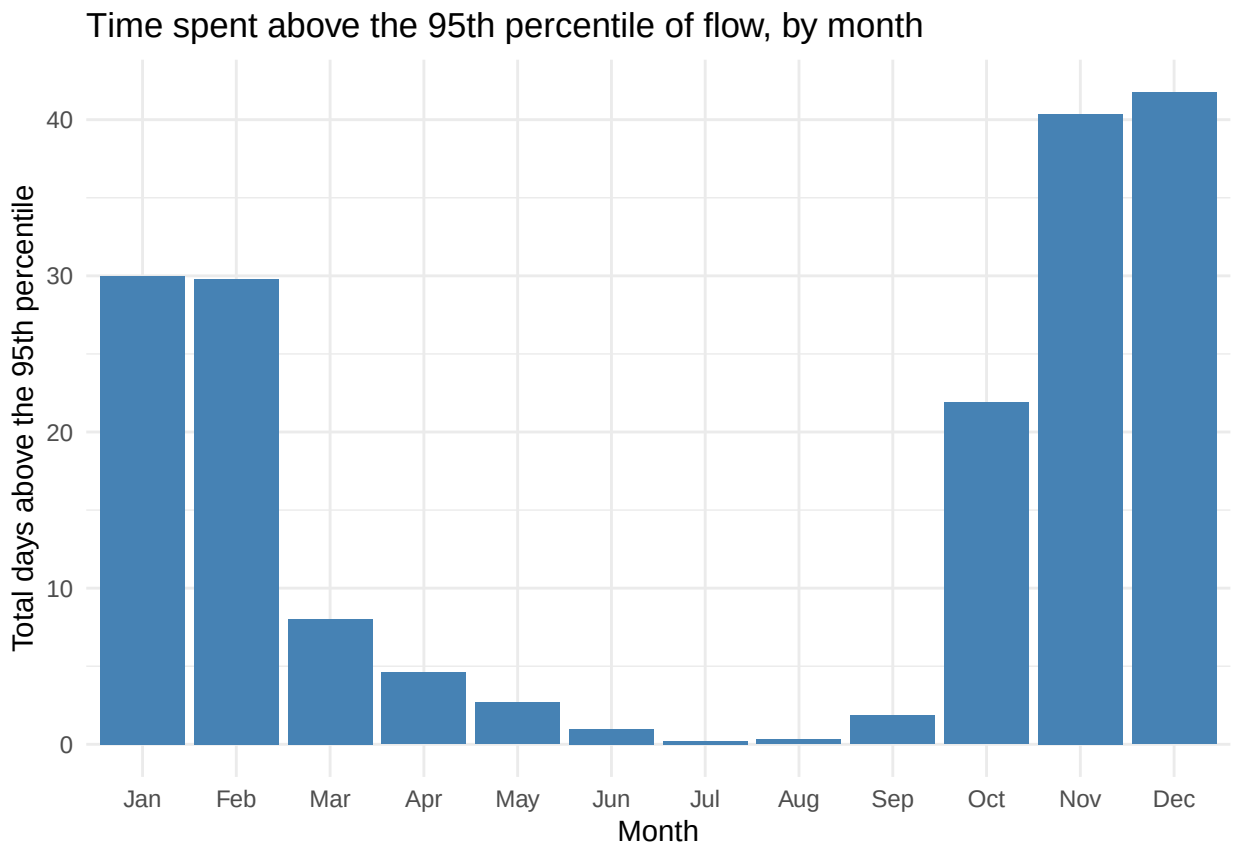
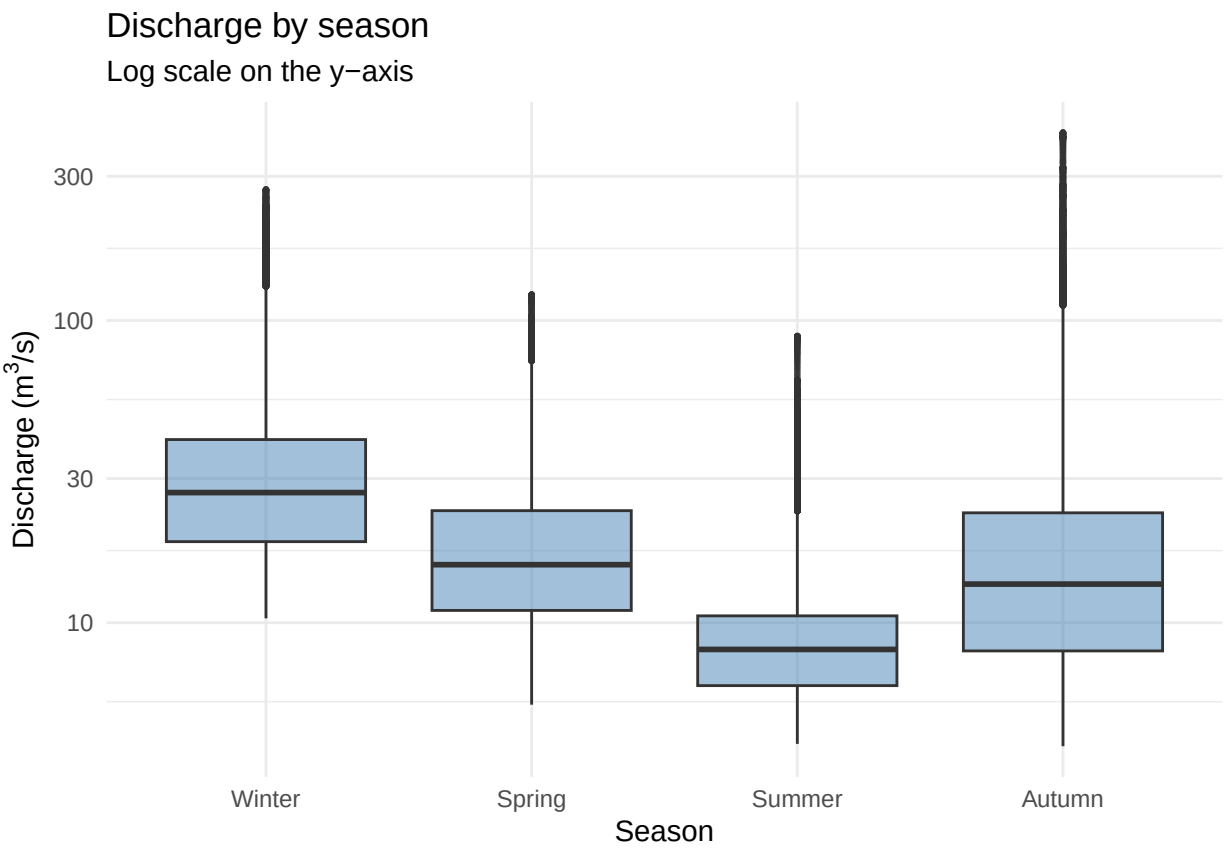


Table 5: Discharge by season (m^3/s)

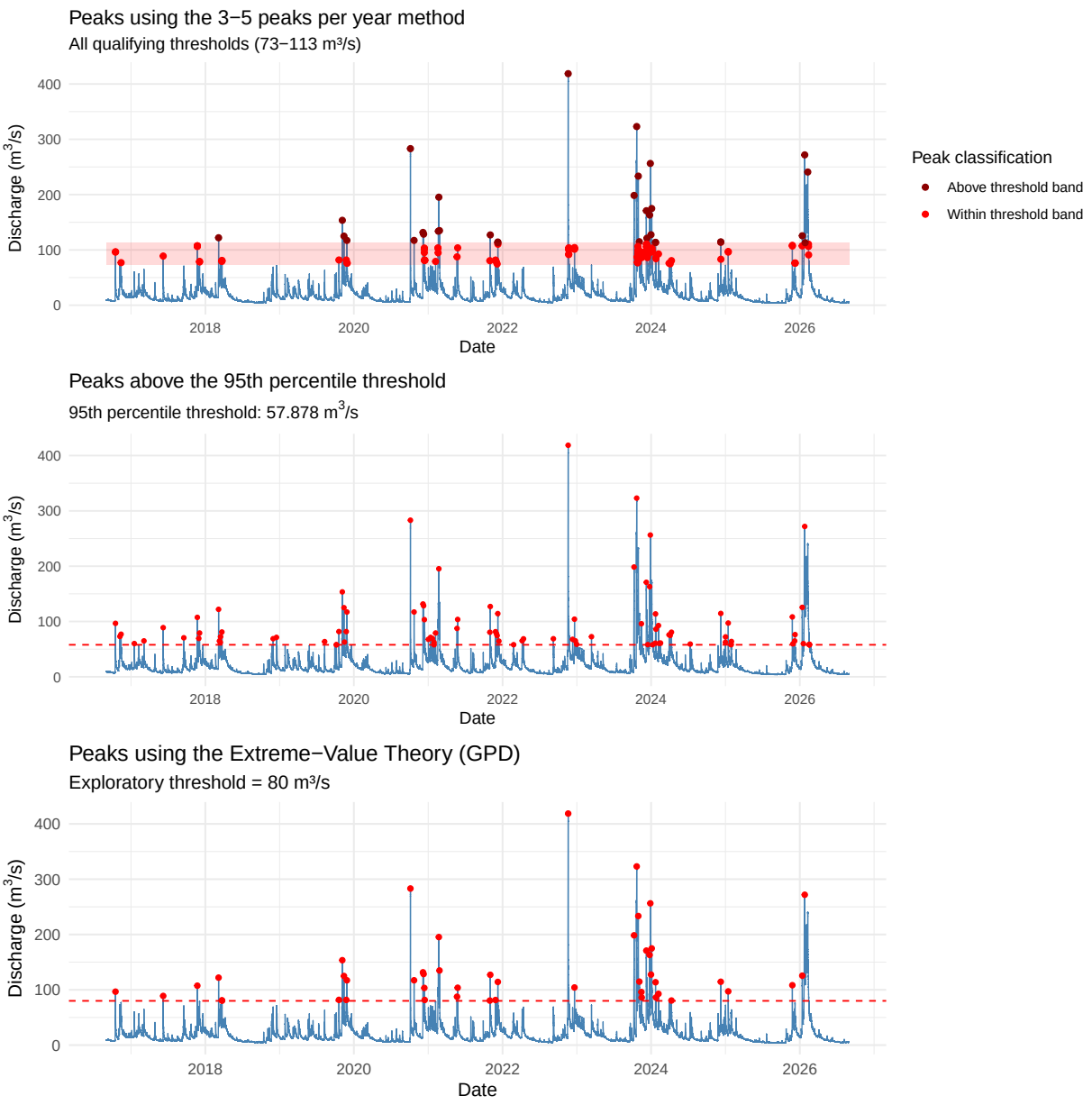
Season	Mean	Median	95th percentile	Maximum
Winter	35.93	26.95	98.58	271.84
Spring	19.00	15.56	40.43	121.98
Summer	9.54	8.16	20.08	88.89
Autumn	22.49	13.42	69.48	418.68



Threshold Selection

The Peaks-Over-Threshold (POT) uses a threshold to extracts the distinct peak flows for extreme events. The threshold is typically selected from three main methods:

1. 3-5 per year method
2. 95th Percentile method
3. Mean residual life threshold analysis



The figures above are a comparison of the thresholds selected by each method. Prior to independence analysis, a practical approach is to choose an initial threshold that is likely to be too low, i.e. ($u = 65$). This means when the distinct extreme events are selected, all the major peaks are being analysed, and subsequentially the threshold can be raised.